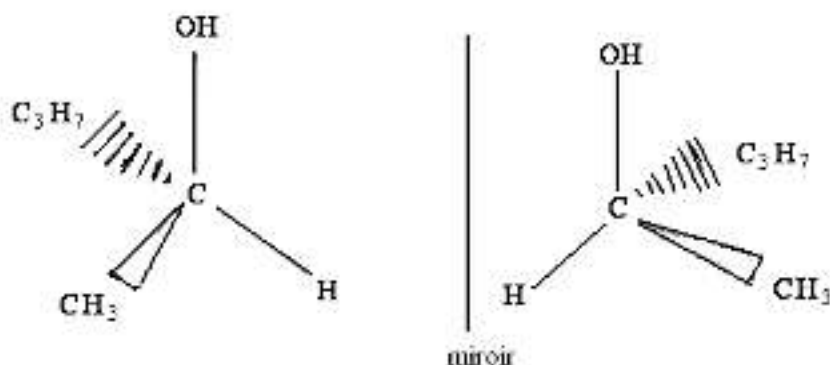
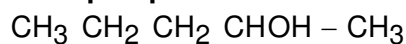
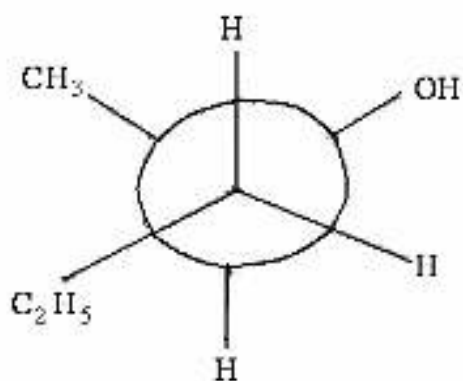


CHIMIE ORGANIQUE

1) a) Représentation en perspective des 2 énantiomères pentan-2ol

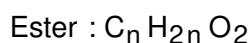


b) Représentation de Newman

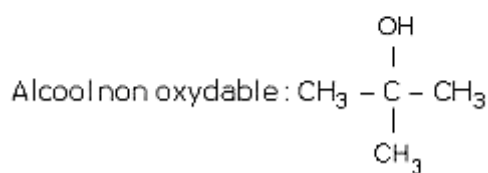
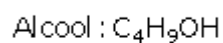
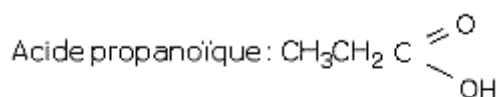
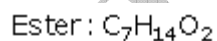


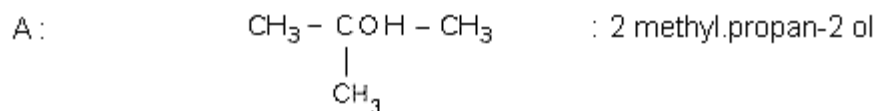
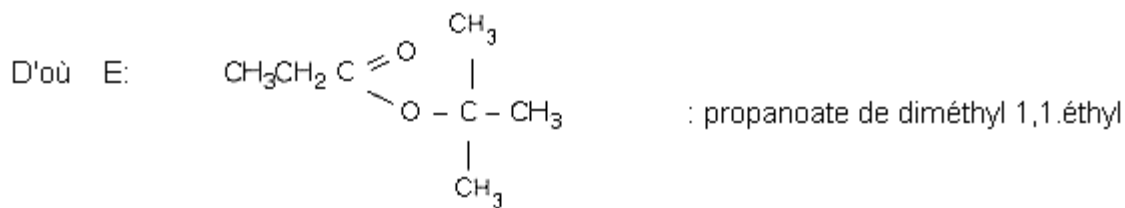
2) a) Formule semi développée de l'ester

$$M_{\text{Ester}} = 130 \text{ g mol}^{-1}$$

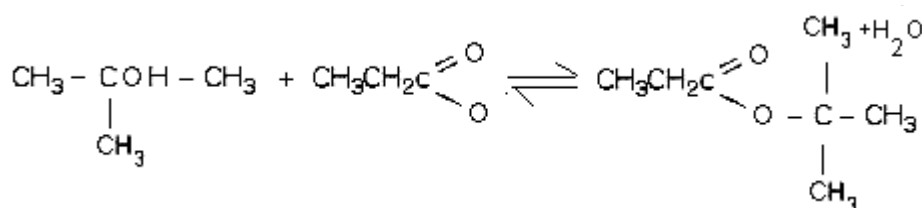
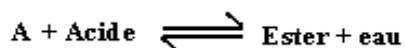


$$14n + 32 = 130 \Rightarrow n = \frac{130 - 32}{14} = 7$$





b) Equation bilan :



CHIMIE GENERALE

1) $\left. \begin{array}{l} \text{pH} = 2,61 \\ -\log C_A = -\log 0,1 = 1 \end{array} \right\} \text{pH} \neq \log C_A$

L'acide benzoïque est faible

2) a- Concentration molaire des espèces chimiques :

Espèces chimiques: $\text{H}_2\text{O}, \text{H}_3\text{O}^+, \text{OH}^-, \text{C}_6\text{H}_5\text{COOH}, \text{C}_6\text{H}_5\text{COO}^-$

$\text{pH} = 2,61 \Rightarrow [\text{H}_3\text{O}^+] = 10^{-2,61} \text{ mol l}^{-1} = 2,45 \cdot 10^{-3} \text{ mol l}^{-1}$

$[\text{OH}^-] = \frac{10^{-14}}{2,45 \cdot 10^{-3}} = 0,408 \cdot 10^{-11} \text{ mol l}^{-1}$

Electroneutralité: $[\text{H}_3\text{O}^+] = [\text{OH}^-] + [\text{C}_6\text{H}_5\text{COO}^-]$

$[\text{OH}^-] \ll [\text{H}_3\text{O}^+] \Rightarrow [\text{H}_3\text{O}^+] \approx [\text{C}_6\text{H}_5\text{COO}^-] \approx 2,45 \cdot 10^{-3} \text{ mol l}^{-1}$

Conservation de la matière :

$C_A = [\text{C}_6\text{H}_5\text{COOH}] + [\text{C}_6\text{H}_5\text{COO}^-]$

$\Rightarrow [\text{C}_6\text{H}_5\text{COOH}] = C_A - [\text{C}_6\text{H}_5\text{COO}^-]$

$= 0,1 - 2,45 \cdot 10^{-3} = 0,0975 \text{ mol l}^{-1}$

b- Calcul de pK_A :

$$pK_A = pH - \log \frac{[C_6 H_5 COO^-]}{[C_6 H_5 COOH]}$$

$$= 2,61 - \log \frac{0,00245}{0,0975} = 4,209$$

$$pK_A = 4,209$$

3) a) Réaction chimique :



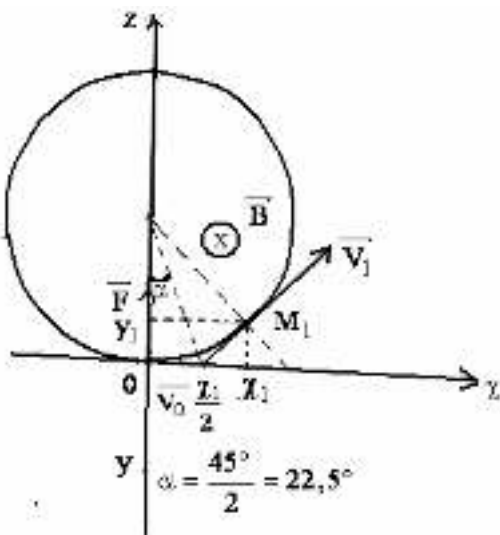
b) Volume de NaOH à l'équivalence :

$$C_A V_A = C_B V_{BE} \Rightarrow V_{BE} = \frac{C_A V_A}{C_B}$$

$$V_{BE} = \frac{0,1 \times 50 \text{ ml}}{0,125} = 40 \text{ ml}$$

ELECTROMAGNETISME

1) Trajectoire du proton :



$$\vec{F} = +e \vec{V}_0 \wedge \vec{B}$$

Rayon de ce trajectoire

$$TCl : m \vec{a} = e \vec{V}_0 \wedge \vec{B}$$

$$m a_N = e V_0 B$$

$$m \frac{V_0^2}{R} = e V_0 B$$

$$R = \frac{m V_0}{e B}$$

$$AN \quad R = \frac{1,7 \cdot 10^{-27} \times 2 \cdot 10^5}{1,6 \cdot 10^{-19} \times 4 \cdot 10^{-2}} \text{ m} = 0,53 \cdot 10^{-1} \text{ m} = 0,053 \text{ m}$$

$$R = 5,3 \text{ cm}$$

3) Calcul de x_1

$$\tan \alpha = \frac{x_1}{2R} \Rightarrow x_1 = 2R \tan \alpha$$

$$\text{AN : } x_1 = 2 \times 0,053 \tan 22,5^\circ = 0,04 \text{ m}$$

$$x_1 = 0,04 \text{ m}$$

$$\text{D'où } x_1 = \frac{4 \times 0,053}{5} \text{ m} = 0,0425 \text{ m}$$

$$\text{B } U(t) = 100\sqrt{2} \sin 100\pi t$$

1) Calcul de R :

$$U = R \cdot I \Rightarrow R = \frac{U}{I} \quad \text{AN } R = \frac{100}{5} \Omega = 20 \Omega$$

2) a) Impédance du circuit :

$$Z = \sqrt{(R + r)^2 + \left(L\omega - \frac{1}{C\omega}\right)^2}$$

$$\text{AN } Z = \sqrt{(20 + 20)^2 + \left(0,1 \times 100\pi - \frac{1}{270 \cdot 10^{-6} \times 100\pi}\right)^2} = 44,549 \Omega \text{ AN}$$

b) Circuit à la résonance :

$$L\omega = \frac{1}{C_2\omega} \Rightarrow C_2 = \frac{1}{L\omega^2}$$

$$\text{AN } C_2 = \frac{1}{0,1 \times (100\pi)^2} = 1,014 \cdot 10^{-4} \text{ F}$$

OPTIQUE

1) Vergence du système :

$$C_1 = \frac{1}{f'_1} = \frac{1}{0,04} \delta = 25 \delta$$

$$C = 25 \delta - 20 \delta = 5 \delta$$

$$C = C_1 + C_2$$

2) a) Position de l'objet AB :

$$\delta = \frac{\overline{OA'}}{\overline{OA}} = 3 \Rightarrow \overline{OA'} = 3 \overline{OA}$$

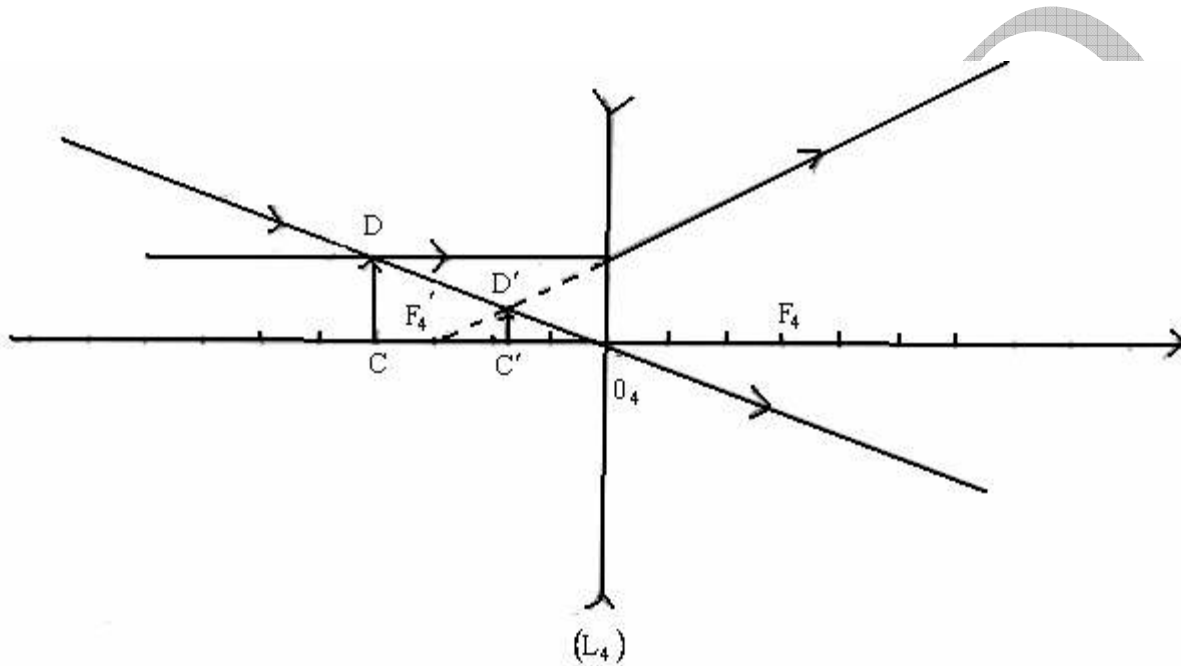
$$\overline{OA} = \frac{\overline{OA'}}{3} = \frac{-12 \text{ cm}}{3} = -4 \text{ cm}$$

b) Calcul de f'_3

$$\frac{1}{f_3'} = \frac{1}{OA'} - \frac{1}{OA} = \frac{OA - OA'}{OA' \times OA} \Rightarrow f_3' = \frac{OA' \times OA}{OA - OA'}$$

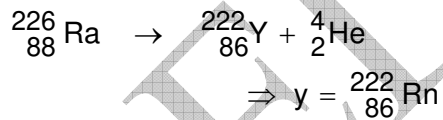
$$f_3' = \frac{-12 \times -4}{-4 + 12} = 6 \text{ cm}$$

3) Construction géométrique : Echelle $\frac{1}{5}$



PHYSIQUE NUCLEAIRE

1) a) Equation de désintégration :



b) Energie par nucléon de Ra

$$\frac{\Delta \varepsilon_l}{A} = \frac{(88m_p + 138m_n - m_{\text{Ra}}) c^2}{226}$$

$$\frac{\Delta \varepsilon_l}{A} = \frac{(88 \times 1,007276 + 138 \times 1,008665 - 225,9771) \times 931,5 \text{ MeV}}{226}$$

$$= 7,66 \text{ MeV par nucléon}$$

2) Calcul de λ :

$$\lambda = \frac{\ln 2}{T} \quad \text{AN} \quad \lambda = \frac{0,69}{3,825 \times 24 \times 3600} \text{ s}^{-1}$$

$$\lambda = 2,087 \cdot 10^{-6} \text{ s}^{-1}$$

PROBLEME DE PHYSIQUE :

A 1° Expression de la vitesse de M en fonction V_0, r, θ

$$\text{T.E.C : } \frac{1}{2} m_1 v^2 - \frac{1}{2} m_1 v_0^2 = -mgh \text{ avec } h = r(1 - \cos \theta)$$

$$v = \sqrt{v_0^2 - 2gr(1 - \cos \theta)}$$

2) Réaction R :

$$\text{T.C.I } \vec{R} + \vec{P} = m \vec{a}$$

$$X'X / R_X + P_X = \frac{m V_M^2}{r}$$

$$R - P \cos \theta = \frac{m}{r} (V_0^2 - 2gr(1 - \cos \theta))$$

$$R = \frac{m}{r} V_0^2 - 2mg + 2mg \cos \theta + mg \cos \theta$$

$$= \frac{m}{r} V_0^2 - 2mg + 3mg \cos \theta$$

$$R_M = \frac{m}{r} V_0^2 + mg(3 \cos \theta - 2)$$

Valeur minimale de V_0 pour que la bille quitte la piste au sommet S.

$$R_S = \frac{m}{r} V_0^2 + mg(3 \cos \theta - 2) \quad (\theta = \pi)$$

$$= \frac{m}{r} V_0^2 - 5mg = m \left[\frac{V_0^2}{r} - 5g \right]$$

$$R_S > 0$$

$$\frac{V_0^2}{r} - 5g > 0 \quad \frac{V_0^2}{r} > 5g \Rightarrow V_0 > \sqrt{5gr}$$

$$\Rightarrow V_{\text{omin}} = \sqrt{5gr}$$

$$V_{\text{omin}} = \sqrt{5 \times 10 \times 0,32} = 4 \text{ ms}^{-1}$$

B 1) a- Moment d'inertie J_1 du cylindre :

$$J_1 = \frac{1}{2} m_1 r^2$$

$$J_1 = \frac{1}{2} \times 0,1 \times (0,04)^2 \text{ kg m}^2 = 8 \cdot 10^{-5} \text{ kg m}^2$$

b- Moment d'inertie J_2 de la tige :

$$J_2 = \frac{1}{12} m_2 R^2$$

The diagram shows a pulley system. A pulley with radius R and moment of inertia Δ is suspended by a rope. A mass m is attached to the rope. Forces shown are tension T , weight P , reaction R , and weight P' .

T.C.I : $M \vec{g} + \vec{T} = M \vec{a}$

$$T = Mg - Ma$$

$$\underbrace{M_{p c / \Delta}}_0 + \underbrace{M_{R / \Delta}}_0 + M_{T / \Delta} = (J_1 + J_2) \theta$$

$$T = T'$$

$$Mg = \left[\frac{J_1 + J_2}{r^2} + M \right] a$$

$$a = \frac{\frac{0,16 \times 10}{8 \cdot 10^{-5} + 1,25 \cdot 10^{-3}}}{(0,04)^2} \text{ m s}^{-2} = 1,61 \text{ m s}^{-2}$$

$$\ddot{\theta} = \frac{a}{r} \quad \text{AN } \ddot{\theta} = \frac{1,61}{0,04} \text{ rad s}^{-2} = 40,25 \text{ rad.s}^{-2}$$